

RotorLab — Rebuild Runbook

Complete, validated rebuild spec for the stirred-tank CFD model — every dimension, build step, and failure mode already discovered, so none of it gets rediscovered. Companion to the RotorLab CFD Guide, which covers the physics.

0. Before You Touch STAR-CCM+

Auto Save (right-click the top-level Simulation node in the tree -> Auto Save): turn it on now. It saves on solver iterations/time-steps, so it protects the MRF and sliding-mesh runs later. It does **nothing** during 3D-CAD or meshing, since nothing is iterating yet.

What actually protects the build phase: manual Save-As at fixed checkpoints, incrementing filenames, saved to a confirmed persistent drive — not assumed local storage on a VDI / cloud desktop.

#	Checkpoint	Save as
1	TankFluid sketch + extrude done	RotorLab_v1.sim
2	Impeller (Shaft+Disc+Blades) united cleanly	RotorLab_v2.sim
3	RotatingFluid / StationaryFluid split complete	RotorLab_v3.sim
4	Volume mesh generated and diagnostics verified	RotorLab_v4.sim

1. Confirmed Geometry Spec — Standard Rushton, T = 0.300 m

Part	Rule	Value
Tank inner diameter, T	—	0.300 m
Sketch 1 circle radius	= T/2	0.150 m
Liquid / tank height, H	= T	0.300 m
Impeller diameter, D	= T/3	0.100 m
Off-bottom clearance, C (disc centerline Z)	floor to impeller centerline	0.100 m
Disc diameter / thickness	= 0.75D / $\approx$ D/50	0.075 m / 0.002 m
Blade length / height / thickness	= D/4 / D/5 / $\approx$ D/50	0.025 / 0.020 / 0.002 m
Number of blades	—	6, at 60°
Shaft diameter	$\approx$ 0.15D	0.015 m
Baffles	count / spacing	4, at 90°
Baffle width / height	= T/10 / full depth	0.030 m / 0.300 m
RotZone diameter — as-spec, do not use	—	0.160 m
RotZone diameter — corrected	radius 0.081 m	0.162 m
RotZone height	—	0.060 m
RotZone Z-center — as-spec, do not use	= C	0.100 m
RotZone Z-center — corrected	offset from disc centerline	0.103 m

The RotZone corrections exist because the as-spec values put its center exactly coincident with the disc's Z-plane, producing a repeatable Boolean dihedral-angle defect (Section 5). Build with the corrected values from the start — do not re-derive from the T/D ratios alone.

2. Build Sequence

2.1 Tank

1) New Simulation -> 3D-CAD Model 1. 2) Sketch 1 on the base plane: circle, radius 0.150 m. 3) Extrude, height 0.300 m, name the body **TankFluid**.

2.2 Every new body from here on

Immediately after creating any primitive or extrude, set **Body Interaction = None** in its properties before doing anything else. It defaults to Merge, All and will silently fuse into TankFluid if left alone.

2.3 Impeller — Shaft, Disc, Blades

4) Shaft: cylinder, diameter 0.015 m, along the tank axis. Body Interaction = None. 5) Disc: cylinder, diameter 0.075 m, thickness 0.002 m, centered at Z = 0.100 m. Body Interaction = None. 6) Blades: 6 rectangular extrudes, length 0.025 m, height 0.020 m, thickness 0.002 m, patterned at 60° around the disc. Place each blade's inner edge **overlapping** the disc's outer radius (0.0375 m) by roughly 1 mm — tangent placement fails the next step. Body Interaction = None on all six. 7) Unite: Target = Disc (or Shaft), Tool Bodies = the other five -> result **Impeller**. Confirm the body count in the tree before continuing.

2.4 Baffles

8) Four wall-mounted baffles, width 0.030 m, full tank height (0.300 m), at 90° spacing. Body Interaction = None on each. Do not attempt to Unite the baffles to each other — they don't touch by design; that failure is expected, not a bug. Leave them as four separate bodies.

2.5 RotZone

9) Cylinder, diameter 0.162 m (radius 0.081 m), height 0.060 m, centered on the impeller axis at **Z = 0.103 m**. Use the corrected values, not the naive T/D-derived ones — see Section 1.

2.6 Fluid Volume Split

10) Subtract: Target = TankFluid, Tool Bodies = [Impeller, Baffle 1–4] -> result **HollowedTank**. 11) Duplicate **both** HollowedTank and RotZone (Ctrl+C / Ctrl+V on each). Both the Intersect and Subtract below consume every body they touch, tool bodies included — RotZone is used by both operations, so it needs its own spare exactly like HollowedTank does. You'll end up with two of each; it doesn't matter which copy goes into which operation. 12) Intersect(HollowedTank, RotZone) -> **RotatingFluid**. 13) Subtract(HollowedTank copy, tool = RotZone copy) -> **StationaryFluid**. There is no dedicated Split tool in this version — duplicate-then-Boolean is the workaround, and it only works if every consumed input has a spare.

3. Parts, Regions, Interface

14) Create Parts from RotatingFluid and StationaryFluid. 15) Assign Parts to Regions (two regions). 16) For each Region, open Properties and confirm the **Parts** field is actually populated — Assign Parts to Regions does not always fill it; link manually if empty. 17) Select one boundary from each region (not the whole regions) and create a **Contact-Mode Boundary Interface**. Confirm the result shows Type = Internal Interface, Topology = In-place, Connectivity = Imprinted. Direct Region Interface (either mode) gives a Heat Exchanger interface instead — wrong type, blocks flow. 18) Rename boundaries: impeller surface -> ImpellerWall, tank wall -> TankWalls.

4. Mesh

19) Create **two separate** Automated Mesh operations — one Input Part each (RotatingFluid, StationaryFluid). The Trimmed Cell Mesher in this version cannot process multiple parts inside one operation.

Setting	Both operations	RotatingFluid operation only
Mesh models	Surface Remesher + Trimmed Cell Mesher + Prism Layer Mesher	same
Base size	0.010 m	same, refine via Target Surface Size
Target / Min surface size	defaults	1.5 mm / 0.5 mm
Prism layers	12	12
Prism stretching ratio	1.2	1.2
Prism total thickness	0.0008 m — Absolute, not Relative	same

20) Generate Volume Mesh. 21) Verify before trusting it: check each operation's own quality summary in the output log (not just the last one printed), then run Mesh Diagnostics on the combined mesh for global min face validity / cell quality / skewness. 22) Section-cut through the tank at mid-height: confirm prism layers are visible on ImpellerWall and TankWalls,

and confirm no visible gap or overlap at the RotatingFluid/StationaryFluid interface — it's non-conformal by construction now that each side was meshed separately.

5. Known Failure Modes

Symptom	Cause	Fix
New body fuses into an existing one on creation	Body Interaction defaults to Merge, All	Set Body Interaction = None immediately on every new body
Unite: "not all tool bodies united"	Bodies only tangent, not overlapping	Give ~1 mm genuine overlap (e.g., blade into disc radius)
Unite fails between baffles	90° apart, never touch — correct behavior	Don't attempt it; use baffles as separate Subtract tool bodies
Connectivity Error: degenerate dihedral angles at a specific Z / radius	RotZone center exactly coincident with disc center plane (Z=0.100)	Offset RotZone: radius 0.081 m, Z-center 0.103 m
"Following Parts need to be assigned to a Region" after Assign Parts to Regions	Region's own Parts property left empty	Manually link Region -> Properties -> Parts
Mesh "finishes" in ~0.00 s, geometry disappears	Automated Mesh's Input Parts or Region's Parts reference a deleted/stale object	Re-link 3 places: Region.Parts, Automated Mesh.Input Parts, Custom Control target
Interface comes in as "Heat Exchanger"	Used Direct Region Interface (either mode) on whole regions	Select individual boundaries -> Contact-Mode Boundary Interface
"Trimmer cannot currently accommodate multiple parts in one AutoMesh Operation"	Trimmed Cell Mesher can't combine 2 parts in 1 operation	Split into 2 separate Automated Mesh operations, one part each
ObjectRegistry / Server Error after deleting an operation	Deleted before confirming the prior step actually succeeded	Never delete-and-see. Check Properties/tree body count first; Undo if it happens
.sim won't open: "corrupted, Invalid begin of file," file is tens of KB	Write truncated — crash, disk full, or non-persistent VDI storage mid-save	Not recoverable. Prevention: confirmed persistent drive + versioned Save-As, Section 0